

Mechatronics Lab Safety Report

Noise floor and rigid body test with DIC

Date:	26/05/2026
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Student contact nr:	0639665173
Supervisor:	Dr Melody Neaves
Lab engineer:	
Head of safety:	Mr Cobus Zietsman

Emergency Contacts:

Contact:	Room nr.	Work nr.	Cell nr.
		-	
Mr C. Zietsman	M212	021 808 4275	-
Campus Security	-	021 808 2333	WhatsApp: 082 808 233
Fire Brigade	-	021 808 8888	-
Ambulance	-	021 883 3444	-

Signatures:

Student:

(Nondumiso Mdabe)

Supervisor:

(Dr Melody Neaves)

Lab Engineer:

(Mr Karl Zapke)

Head of Safety:

(C Zietsman)

Pressure vessels or pipes (check relevant box):

- ☒ No pressure vessels or pipes with pressure in excess of 50kPa are involved in this project.
☐ Pressure vessels or pipes in excess of 50kPa are involved – additional signature and report required (refer to Safety Report Guidelines on SUNLearn).

Hot work / working at heights / confined entry / excavation (check relevant box):

- ☒ No hot work / working heights / confined entry / excavation work involved in this project.
☐ Hot work / working heights / confined entry / excavation work (underline relevant work type(s)) involved in this project – additional signature and report required (refer to Safety Report Guidelines on SUNLearn).

Overview of Testing

Type of test and standard (if applicable):

Test type	Noise-floor and rigid body translation test with DIC
Standard(s)	iDIC good practices guide

Detailed Experimental Procedure

Table 1: Experiment materials

Component	Description	Function
Aluminium test specimen	100mm x 80mm x1.6mm flat plate	Provide rigid and non-deforming surface ideal for noise floor test
Matte black spray paint	Aerosol coating	Creates the high contrast speckle pattern
Mounting platform	Rigid fixture	Ensure that the test specimen stays fixed with zero displacement
Calibration plate	80mm x 80mm	Used to calibrate DIC system before testing
Safety mask	PPE	Prevent inhaling spray paint

Table 2: Equipment specification

Equipment	Specification	Function
LaVision USB-3 Imager E-Lite Camera	Resolution: 2464 x 2056 pixels; 8-bit grayscale	Capture images for DIC calculations
LaVision 35mm C-mount lens	35mm focal length	Focuses image at working distance ($\approx 350\text{mm}$)
LaVision LED linear array	20W max power, adjustable intensity	Provides illumination across the specimen's surface
Commercial DIC calibration plate	80mm x 80mm high-contrast dot pattern custom board	Calibrate camera for distortion correction
Computer	Intel core i5 processor and 16GB RAM	Runs the DIC software and process image data

Experiment 1: Noise floor test

Aim

The aim of the experiment is to quantify the noise floor of the of the DIC setup and demonstrate that adaptive closed loop lighting system introduces less artificial strain than static illumination.

Procedure

Static illumination test

1. Switch on the static LED illumination system
2. Adjust to achieve acceptable image contrast. Ensure there is no over saturation or under exposure.
3. Capture 30 images at 1 second intervals.
4. Store images in a dedicated folder.

Adaptive illumination test

1. Switch off the static illumination system.
2. Switch on the adaptive illumination system.
3. The control system should adjust the LED light intensity in real time to optimise image quality based on a selected metric.
4. Capture 30 images at 1 second intervals.
5. Store images in a dedicated folder for analysis.

Experiment 2: Ambient lighting

Aim

The aim of the experiment is to reach similar quality score independent of ambient light. To evaluate whether the adaptive illumination system under can reproduce image quality under varying ambient lighting conditions.

Procedure

1. Set the static LED illumination to a fixed baseline intensity.
2. Start the adaptive illumination system and allow it to settle.
3. Monitor and store the MIG score, entropy value and the PWM output.
4. Repeat the steps above with a different intensity level.

Performance is evaluated based the system's ability to maintain consistent MIG values despite changes in ambient light

Warning Symbols

The following symbols are applicable to this setup.



(a) Electrical hazard



(b) General warning

Figure 1: Applicable warning/hazard symbols

General Laboratory Safety

The following general laboratory safety instructions are applicable:

- No afterhours testing may be performed without the necessary permissions¹.
- Full supervised training is required before testing may be undertaken. Permission to proceed with unsupervised testing should be signed off by the lab engineer.
- Closed shoes must be worn at all times.
- Emergency equipment must be located and easily accessible.
- Students may not work alone in the laboratory.
- Emergency exits must be known. The nearest exits applicable to the setup are provided in Appendix A of this document
- Loose clothing may not be worn. Loose hair must be tied up.
- Good housekeeping practices should be maintained during testing. The lab should be completely clean, including all equipment stored away, after testing. Refer to the General Housekeeping section for particulars regarding practices to be followed for this specific setup.
- No food or drink is permitted in the laboratory.
- Safety report must be visible and accessible during testing.
- No equipment or test may be left unattended.

Anticipated Interactions with other Laboratory Users

General Housekeeping

The following housekeeping steps must be taken:

- Return any tools to the allocated slots.
- Report any damaged or broken components, tools or machinery.
- Ensure the workspace is clear and clean.
- Ensure no personal belongings are left behind.

Fire Safety

The testing does not include any direct fire risk. In the event of a fire, the evacuation plan seen in Appendix A should followed with the nearest exits circled in red. No earphones should be worn while testing so that the fire alarm can be heard.

Activity Based Risk Assessment

The following risks and mitigating steps are associated with certain activities associated with this experimental setup procedure:

Activity	Risk	Risk Type* (P/E)	Mitigating Steps	Classification of Risk Severity
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¹ Written permission from supervisor and approved by the chief safety officer. Attach proof of this permission to this document and note here the times and conditions of the arrangement.

General				
Entering the lab	Injuries to hands from the gates	P	Use caution when using the gates	Acceptable
Moving around in the lab	Tripping or knocking equipment over	P	Never use a cell phone when moving around and be aware of surroundings	Acceptable
Moving any DIC equipment out of the way	Dropping expensive DIC hardware	E	Clear a path to the MTS before starting to move the equipment and ask someone to assist with carrying it into position	Possible
Power outages	Damage to equipment or loss of data	E	Be ware of load shedding schedules and be sure to switch all equipment	Possible
Personal valuables in the laboratory	Theft of valuables from the laboratory	P	Do not bring unnecessary valuable items to laboratory sessions. Valuables that are brought to the laboratory should be placed in a safe and visual location	Acceptable risk
Tidying lab	Tripping	P/E	Do not trip over cables	Possible risk
	Cuts	P	Beware of sharp edges on the compact specimen	Possible risk
Constructing system				
Soldering	Cause a fire	P/E		Possible risk
	Burn skin	P		Possible risk
Cutting wires	Cutting fingers	P		Possible risk
	Cutting other components	P		Acceptable
Specimen prepping				
Spray paint	contact with skin or eyes	P	Wear protective glass and gloves. Prevent contact with exposed skin	Possible risk
	Inhalation of fumes or respiratory irritation	P	Put on mask to prevent inhaling fumes	Possible risk
	Paint overspray on lab surfaces or equipment	E	Ensure that there is no equipment near spray booth. If any of the surfaces are sprayed, then notify the lab manager	Possible risk

Experiments				
Flashing light	Photosensitive epilepsy	P	Warning labels and announcement	Possible risk
High intensity light	Eye strain or damage	P	Diffuse light and warning labels	Acceptable risk
	LED overheating and damage	E	Use heat sinks and current limiting driver	Possible risk
Adjusting focus and lens	Fingerprints Or smudges on the lens	E	Use lens cloth and avoid touching lens surfaces	Low risk
Mounting the test specimen	Specimen falling or shifting position	E	Secure specimen with proper clamps and mounts	Possible risk
Running high speed DIC capture	Data loss due software/hardware crash	P	Reduce speed capture, save data regularly and use backup storage	Possible risk
Processing data	Data corruption due to file handling errors	P	Use version control and store backups	Possible risk

*P – personal, E - equipment

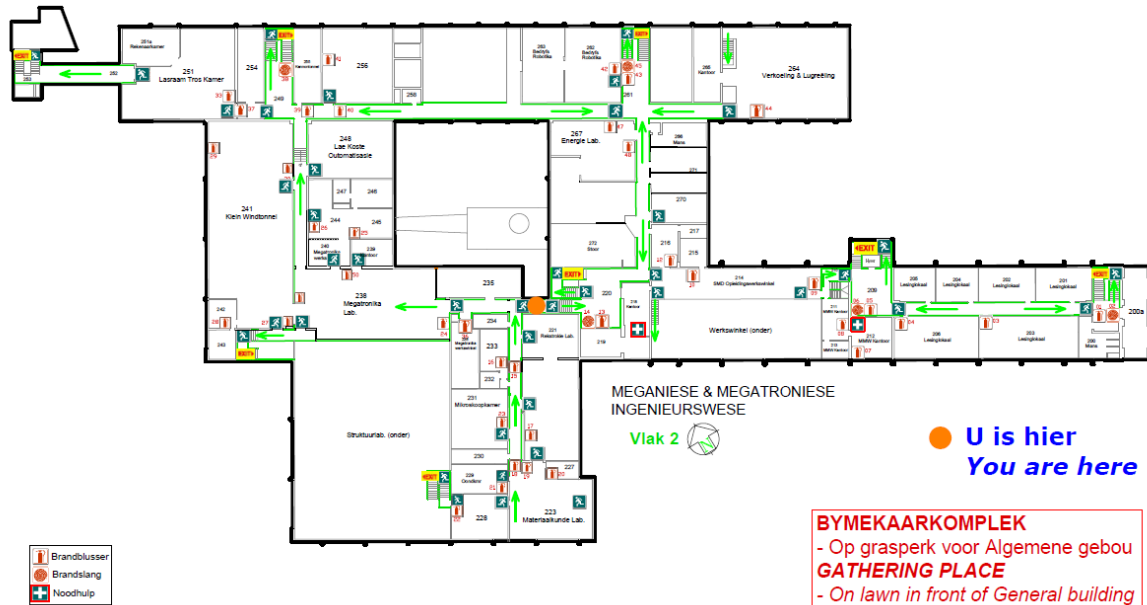
Disciplinary Actions

Failure to comply with any of the aforementioned safety regulations or procedures will result in disciplinary action. Students will be issued an initial warning: after three warnings, the lab access is revoked for a month.

Appendix A: Emergency Evacuation Plans

ONTSNAPPINGSROETE

ESCAPE ROUTE



ONTSNAPPINGSROETE

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